Algebra II with Trigonometry Chapter 11.4

Olympiad Problems (GPT-5.4)

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Problems

1. Find the center and classify the conic:

$$4x^2 + 9y^2 - 24x + 54y + 36 = 0.$$

2. A parabola has focus $(5, -1)$ and directrix $x = -1$. Without writing the full equation first, find its vertex and the value of y when $x = 8$.

3. An ellipse has center $(2, 1)$, passes through $(2, 5)$, and has foci $(2, -2)$ and $(2, 4)$. Find its equation.

4. A hyperbola has asymptotes

$$y - 3 = \pm 2(x + 1),$$

and one vertex at $(1, 3)$. Find its equation.

5. Determine all real values of k for which

$$x^2 - 4x + y^2 + 2ky + 4k - 5 = 0$$

represents exactly one point.

6. A conic is given by

$$9x^2 - 16y^2 - 54x - 64y + 19 = 0.$$

Find its center, identify the conic, and determine its asymptotes mentally by transforming it to standard form.